



ADJUSTABLE GUARD STORAGE VEHICLE

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

ADJUSTABLE GUARD STORAGE VEHICLE	0
1.0 Introduction.....	1
2.0 Best Practice Description	2
3.0 Implementation Steps.....	4
4.0 Benefits	7
5.0 Resources Required.....	7
6.0 Supporting Attachments / References	7
7.0 Related Best Practices	7
8.0 Acknowledgements	8

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1.0 Introduction

Following the disassembly of excavators and dozers, the resulting guard pieces are not properly disposed in a storage area of the workshop. As such, the repair process can be delayed and inefficient due to parts being unevenly scattered around multiple pallets.

To overcome this issue, a guard storage vehicle was designed with a vertical partition feature. This feature will then allow the guard parts to be stored altogether and to occupy less space. Another design feature is the adjustability of the vehicle in the sense that technicians can customize the gaps between iron partitions according to their needs.

Overall, there will be an increase in efficiency and simplicity; the relevant parts will be compartmentalized and transported simultaneously to the washing and measurement department.



BEFORE

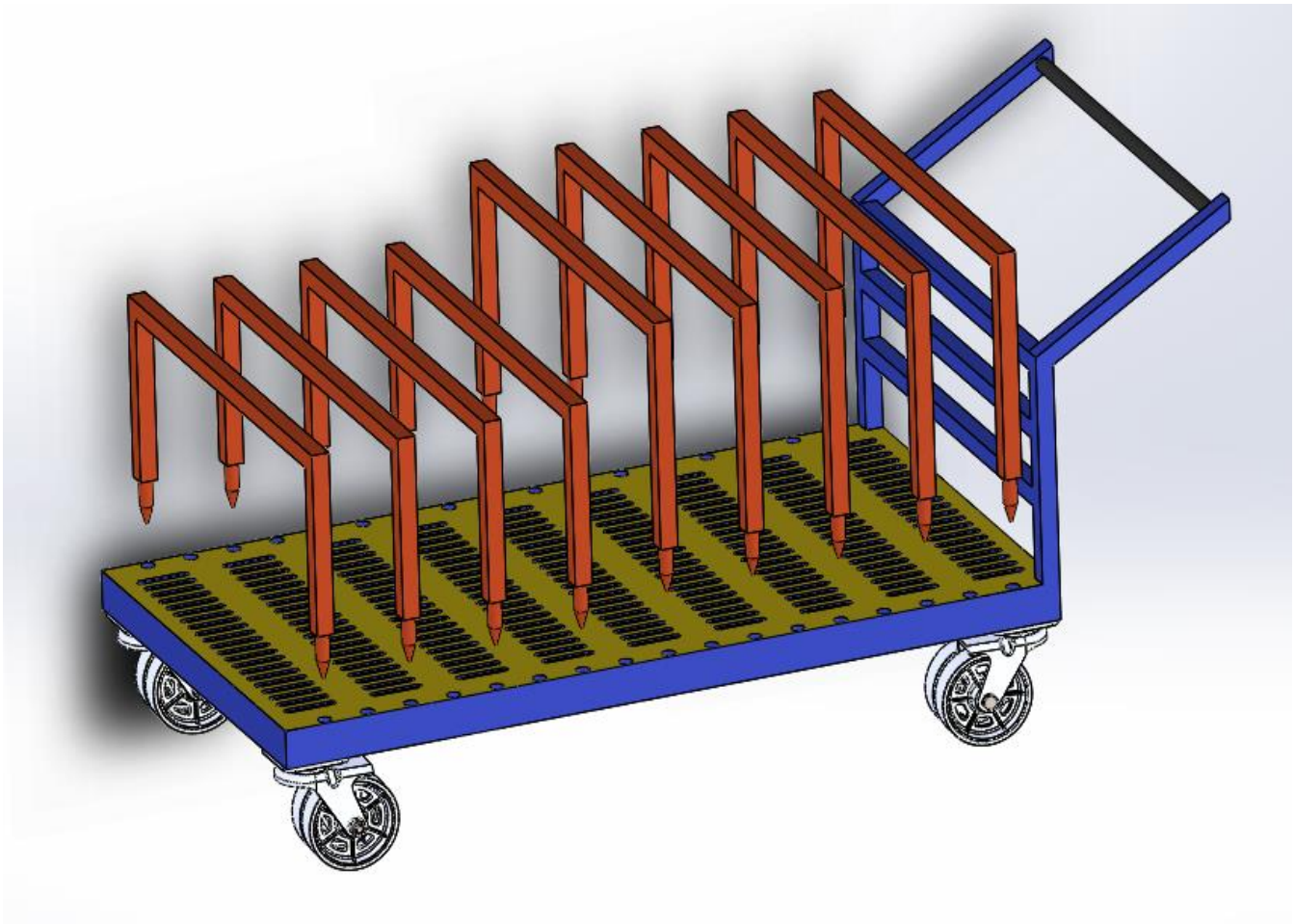


AFTER

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2.0 Best Practice Description

To build the designed vehicle, the welding department mounted the wheels under a sheet plate to provide mobility, and added partitions on top of the plate at specific intervals. Furthermore, the intervals between the partitions were designed to be adjustable as needed by the technicians. A handle was also attached to facilitate directions and usage. By using this vehicle, storing the guard parts has become more organized while transitioning between departments has become easier.



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3.0 Implementation Steps

1- Wheel and iron partition placement



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2- Attachment of iron partitions on top of the sheet plate

FIXED



ADJUSTABLE



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3- Attachment of the handles



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4.0 Benefits

- 1- Improvement in process speed and efficiency
- 2- Improvement in process flow due to parts' compartmentalization (transporting, washing, removing-attaching process)
- 3- Prevention of irregular placement of the guard parts. (The extra pallet occupation was removed)
- 4- Gaps' customization in between iron partitions as needed by the technicians.

5.0 Resources Required

A vehicle can be built with 2 technicians working for 1 day.

Piece	Measurement	Quantity	Unit Cost (\$)
Plate	(1mx2mx3mm)	1	20
Iron Partitions	(40cmx40cmx2mm)	10	4
Iron pipe	(1mx2mmx2mm)	5	5
Wheel	-	2(locked)+2(regular)	20

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgments

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